SENG 448 Engineering Design

Software Requirements Specification (SRS)

Smart Personnel Newspaper System

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1. Introduction

1.1 Purpose of the System

The purpose of the Smart Personnel Newspaper System is to collect news from multiple media sources and present them in a personalized newspaper format for users. The system aims to help users access relevant information quickly by organizing news according to their interests and preferences.

The system gathers news from different media outlets, categorizes them, and generates a personalized news interface. Instead of displaying full news articles from all sources, the system provides references and links to the original media outlets so that users can access the news directly from the source.

The system shall also display additional information such as weather updates and stock market data on the main interface. Furthermore, the system can identify similar news events reported by different media sources and present the related sources to the user.

1.2 Scope of the System

The Smart Personnel Newspaper System is designed to collect news from multiple internal and external sources, classify the collected content into categories, and generate a personalized newspaper for users based on their interests and preferences.

The system supports news collection, categorization, personalized newspaper generation, search and filtering, event and announcement management, and printable output.

The system may also support multilingual news sources from different media outlets. In addition to personalized news, the system can present additional information such as weather updates and stock market information on the main interface.

Furthermore, the system can identify similar news events reported by different media outlets. Instead of displaying the full content from each source, the system presents the related media sources and allows users to access the original news directly through external links.

1.3 Stakeholders

The main stakeholders of the Smart Personnel Newspaper System include administrators, end users, content managers, and system managers.

Administrators are responsible for managing the system configuration, news sources, and overall system functionality.

End users interact with the system to view personalized news, search for specific topics, and access news sources through external links.

Content managers are responsible for reviewing news categories, organizing content, and ensuring that information presented in the system is relevant and properly categorized.

System managers or organizational supervisors may monitor how the system distributes information and ensures that the system operates properly.

2. System Overview

The Smart Personnel Newspaper System is designed to collect news from multiple media sources and organize them into a personalized newspaper format for users. The system gathers information from different external news providers and categorizes the collected content to improve accessibility and usability.

Users can browse personalized news, search for specific topics, and filter news according to their interests. Instead of displaying full news articles within the system, the platform provides links that allow users to access the articles directly from their original news sources. Each news item includes a reference link that redirects users to the corresponding news source for detailed reading.

The system may also support multilingual news sources in order to provide access to information from different regions and languages. In addition to personalized news content, the system homepage can display supplementary information such as weather updates or financial indicators to improve the user experience.

Furthermore, the system can analyze collected news to detect similar news events reported by different media outlets. When similar news is identified, the system groups the related sources together and provides links that redirect users to the corresponding news sources. The system can also recommend related news articles from the same or similar categories, helping users discover additional relevant content.

3. User Types

3.1 Administrator

Administrators manage and maintain the overall operation of the system. They are responsible for configuring system settings, managing news sources, maintaining user accounts, and monitoring system performance. Administrators also ensure that the system operates correctly and efficiently.

3.2 Personnel (End Users)

Personnel are the primary users of the system. They access the platform to read personalized news and announcements based on their interests. Users can browse categorized news, search for specific topics, filter news content, and access links that redirect them to the original news sources for detailed information.

3.3 Content Manager

Content managers supervise the news content within the system. They review collected news, organize news categories, and ensure that announcements and news information are properly classified. Content managers may also update or edit news categories when necessary.

3.4 Organization Management

Organization management represents decision-makers who use the system to follow important announcements, events, and news related to the organization. They benefit from receiving organized and relevant information that helps them stay informed about institutional updates.

4. Functional Requirements

4.1 News Data Collection Module

Purpose: *Automatically collect news from multiple sources.*

FR01: The system shall automatically collect news articles and their metadata (title, publication date, and source) from online newspapers, news portals, and blogs using web crawling techniques.

FR02: The system shall retrieve news data from supported external news providers through API integration when APIs are available.

FR03: The system shall fetch news data at predefined and configurable intervals such as hourly or daily.

FR04: The system shall detect and remove duplicate news articles based on attributes such as title, source, URL, or content similarity during the data collection process.

4.2 News Classification & Categorization Module

Purpose: *Classify collected news according to user interests.*

FR05: The system shall automatically analyze news text from sources and assign it to predefined categories (e.g., Technology, Politics, Sports, Economy).

FR06: The system shall support multi-label classification allowing a single news article to be assigned to multiple categories if the content pertains to more than one field.

FR07: The system shall automatically extract and tag representative keywords from each news article to facilitate better indexing and discovery.

FR08: The system shall automatically generate concise summaries (1–2 sentences) for each news article using AI-based text processing techniques.

FR09: The system shall automatically detect and tag important entities such as persons, organizations, locations, and dates within news articles using AI-based content analysis.

4.3 Personal Newspaper Generation Module

Purpose: *Generate a customized daily newspaper for each user.*

FR10: The system shall generate a personalized newspaper for each user based on the user's selected interests and preferred news categories.

FR11: The system shall allow users to filter and organize news articles in their personalized newspaper based on date, source, popularity, and relevance.

FR12: The system shall display each news article together with its source information and publication date and generate daily or weekly personalized newspaper editions in a digital newspaper layout format.

4.4 Hardcopy Printing Module

Purpose: *Allow users to print their personalized newspaper.*

FR13: The system shall export the user's personalized newspaper into a high-fidelity, print-ready PDF format.

FR14: The system shall provide a print preview interface and allow users to choose layout templates such as A4, Tabloid, or Booklet before downloading or printing.

4.5 Event & Announcement Module

Purpose: *Notify users about important institutional events.*

FR15: The system shall automatically detect and extract important events and announcements from news articles using AI-based content analysis and display them in the event and announcement section.

FR16: The system shall categorize detected events into predefined categories (academic, social, deadlines, exams) and allow users to view event details including date, location, and description.

4.6 Search & Filtering Module

Purpose: *Provide advanced content discovery features.*

FR17: The system shall allow users to search for news articles using keywords and display relevant results.

FR18: The system shall allow users to filter search results based on category, date, source, or popularity.

FR19: The system shall allow users to bookmark or save news articles for later reading and highlight trending or most searched topics.

4.7 Analytics & Recommendation Module

Purpose: *Improve personalization through intelligent analysis.*

FR20: The system shall allow users to select and update their preferred news categories for personalization purposes.

FR21: The system shall track and record user reading behavior and interaction history and recommend news articles based on previous interactions.

FR22: The system shall identify trending topics and generate engagement statistics such as most viewed news and popular categories.

FR23: The system shall rank and display news content adaptively based on user interests, interaction data, and recommendation results.

4.8 Reporting & Administration Module

Purpose: *Provide administrative control and system monitoring.*

FR24: The system shall allow administrators to add, update, and remove news sources and news categories and manage user roles and permissions.

FR25: The system shall record system activities and errors in log files and generate usage reports including user activity statistics and news access statistics.

4.9 Archive & History Module

Purpose: *To store and manage previously generated personalized newspaper editions and allow users to access, review, and search past editions when needed.*

FR26: The system shall maintain an archive of previously generated personalized newspaper editions for each user.

FR27: The system shall allow users to access, view, and download past personalized newspaper editions.

FR28: The system shall allow users to search within archived personalized newspaper editions.

4.10 News Source / Publisher Profile Module

Purpose: *Represent news sources as visible publisher profiles within the system and allow users to explore news content through publisher-based pages.*

FR29: The system shall represent each news source as a visible publisher profile within the application or website interface.

FR30: The system shall display publisher profile information such as source name, logo, description, and category where available.

FR31: The system shall provide a dedicated publisher page for each news source to display its related news articles.

FR32: The system shall allow users to browse and access news articles through publisher profiles within the system interface.

FR33: The system shall organize publisher profiles in a discoverable section where users can explore available news sources.

FR34: The system shall allow administrators to manage publisher profile information such as logo, description, and display settings.

FR35: The system shall allow users to follow or subscribe to publisher profiles in order to receive updates from their preferred news sources.

FR36: The system shall allow users to like or react to news articles displayed on publisher profile pages.

5. Non-Functional Requirements

NFR1: The system shall process user interaction data and generate personalized news recommendations with acceptable response time.

NFR2: The system shall protect user interaction data and personal preference information from unauthorized access.

NFR3: The system shall support multiple users simultaneously while maintaining stable performance during analytics and recommendation processes.

NFR5.1: The system shall present events and announcements through a clear and user-friendly interface that allows users to easily view upcoming events and announcements.

NFR5.2: The system shall present event and announcement content in a responsive layout that adapts to different screen sizes and devices.

NFR5.3: The system shall ensure that event notifications and reminders are delivered to users without duplication or loss.

NFR5.4: The system shall ensure that event and announcement information remains accurate and updated whenever changes are made by administrators.

6. System Constraints

This section defines the practical engineering limitations that affect the development and operation of the Smart Personnel Newspaper System. These constraints are related to budget, hardware resources, computing capacity, and technology choices.

6.1 Budget Constraints

The Smart Personnel Newspaper System shall be developed with a minimal budget, prioritizing cost-effective solutions throughout the entire development lifecycle.

The system shall be developed primarily using open-source tools, libraries, and frameworks in order to minimize development costs.

The project shall avoid the use of commercial or paid software licenses whenever possible.

Free-tier API services shall be preferred for news data collection and integration where applicable.

Cloud hosting and storage solutions shall be selected based on free or low-cost plans available for academic and non-commercial use.

6.2 Hardware Constraints

The system shall be designed to operate on standard mobile devices as the primary hardware target. No specialized or high-end hardware shall be required.

The system shall support the following minimum mobile hardware requirements:

Minimum 2 GB RAM for mobile devices.

Minimum 100 MB of free storage for application installation; additional space may be required for cached news content.

A front-facing display with a minimum resolution of 360×640 pixels shall be supported.

A stable internet connection (Wi-Fi or mobile data) shall be required for news data collection and real-time updates.

For the secondary web platform, the system shall run on any standard personal computer capable of running a modern web browser, without specialized hardware requirements.

6.3 Memory Constraints

The system shall manage its memory usage efficiently to ensure stable and responsive operation on target hardware environments.

The system shall manage memory efficiently and avoid unnecessary memory consumption.

The system shall operate within the memory limits of typical desktop environments.

News data caching mechanisms shall be optimized to prevent excessive RAM usage during bulk content fetching and classification operations.

PDF generation for hardcopy printing shall release allocated memory promptly after the export process is completed.

6.4 CPU Constraints

The system shall be optimized to keep CPU usage at acceptable levels during normal operation, ensuring a smooth user experience even on lower-end hardware.

The system shall ensure that CPU usage remains within reasonable limits during normal system operation.

The system shall avoid computationally expensive operations that could degrade system performance.

AI-based classification and recommendation tasks shall be executed asynchronously or during off-peak periods to prevent UI blocking.

Automated web crawling and news fetching operations shall be throttled to avoid excessive CPU load during scheduled data collection tasks.

6.5 Platform Constraints

The system shall primarily target mobile platforms, with secondary support for web browsers. The mobile application shall serve as the main interaction channel, while the web interface shall provide equivalent core functionality for desktop users.

The system shall support the following mobile platforms as primary targets:

Android (Android 10 and above)

iOS (iOS 14 and above)

The system shall additionally support the following web browsers as a secondary platform:

Google Chrome (version 110 and above)

Mozilla Firefox (version 110 and above)

Microsoft Edge (version 110 and above)

The system's user interface shall be responsive and shall not require browser extensions or plugins for core functionality.

6.6 Technology Constraints

The system shall be implemented using widely adopted and well-supported technologies to ensure long-term maintainability, developer accessibility, and alignment with open-source budget constraints.

The system shall be implemented using widely supported programming languages and development frameworks to ensure maintainability and long-term support.

Backend services shall be developed using Python or Node.js, both of which are open-source and have large, active developer communities.

The mobile application shall be developed using React Native with Expo framework, enabling a single shared codebase for both iOS and Android platforms, as well as secondary web support via Expo Web.

The database layer shall use an open-source relational or document-oriented database such as PostgreSQL or MongoDB.

AI-based classification and recommendation modules shall utilize open-source machine learning libraries such as scikit-learn, TensorFlow, or Hugging Face Transformers.

All third-party libraries and dependencies used in the project shall be licensed under permissive open-source licenses (e.g., MIT, Apache 2.0).